



Department of Information Technology
Final Year Project Topic Submission

Title:

Cybersecurity Intrusion Detection using Generative Adversarial Networks (GANs)

Description:

The project explores the application of Generative Adversarial Networks (GANs) in enhancing cybersecurity intrusion detection systems. Traditional intrusion detection systems (IDS) face challenges in accurately identifying novel and sophisticated cyber threats due to their dependency on labelled data and predefined signatures. GANs, which consist of a generator and a discriminator, can be utilized to simulate realistic attack scenarios, generating synthetic data that mimics potential cybersecurity threats. This synthetic data can be used to train more robust IDS models, improving their ability to detect and respond to a wider range of intrusions, including zero-day attacks.

Existing Methodologies / Applications:

1. **Traditional Intrusion Detection Systems (IDS):**

These systems primarily rely on signature-based or anomaly-based detection. Signature-based systems use predefined patterns of known threats, while anomaly-based systems detect deviations from normal behavior. However, they struggle with detecting novel attacks and require extensive labeled data for training.

2. **Machine Learning-Based IDS:**

Recent advancements have incorporated machine learning techniques into IDS, using supervised learning methods to classify network traffic as benign or malicious. However, these methods still face challenges with data scarcity and the detection of sophisticated attacks.

3. **GAN-based Approaches:**

- **Synthetic Data Generation:** GANs have been applied to generate synthetic attack data, providing a richer dataset for training IDS models. This approach improves the generalization capabilities of the models, allowing them to detect previously unseen attacks.
- **Adversarial Training:** By simulating adversarial scenarios, GANs help in creating robust IDS that can withstand and identify attacks that attempt to evade detection.

4. **Applications in IoT and Smart Environments:**

GANs have been used in IoT networks to detect intrusions by creating synthetic datasets that simulate attacks on IoT devices, improving the security of these increasingly prevalent networks.



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Features/ Scope:

- **Enhanced Detection Accuracy:** By integrating GANs into IDS, the project aims to significantly improve the accuracy of detecting both known and unknown cyber threats.
- **Real-time Intrusion Detection:** The system will be designed to operate in real-time, providing immediate responses to potential intrusions.
- **Adversarial Training and Resilience:** The IDS will be trained using adversarial scenarios, making it more resilient to sophisticated attacks that traditional systems might miss.
- **Applicability to Various Environments:** The project will explore the application of the GAN-based IDS in different environments, including traditional networks, IoT systems, and smart environments.

Dataset:

The datasets used in this project will include publicly available cybersecurity datasets such as the KDD CUP 99, NSL-KDD, or CICIDS 2017. Additionally, GANs will generate synthetic datasets to simulate various types of cyber-attacks, enriching the training data and helping in the detection of complex, previously unseen attacks.

References:

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Project group Details:

60003210017 I1 Shreya Parkar

SAP ID Division Name Sign

SAP ID Division Name Sign

Guide Name

Signature with date



Shri Vile Parle Kelavani Mandal's

DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING

(Autonomous College Affiliated to the University of Mumbai)

NAAC Accredited with "A" Grade (CGPA : 3.18)



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